FABRIZIO RUSSO

Research Fellow, Imperial College London

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SUMMARY

Causal learning researcher with expertise spanning causality, machine learning, applied statistics, and formal AI methods. I hold an MSc in Applied Statistics and a PhD in Safe and Trusted AI and bring six years of professional experience applying statistical methods in regulated financial services. Alongside my research, I teach statistical and causal learning, supervise student research projects, represent research staff and contribute to departmental EDI initiatives. I am now building an independent academic profile that connects causal learning with substantive problems in healthcare, finance and public policy.

WORK EXPERIENCE

Imperial College London

- Research Fellow (AI) Imperial Policy Forum — Full-time: Sep 2026-present — UK
- Research Associate — Full-time: Mar 2025-Aug 2026 — UK
  - ERC Proof of Concept grant: Causal Argumentative Learning Assistant (CArLA)
- Research Assistant — Full-time: Jan 2025-Feb 2025 — UK
- Graduate Teaching Assistant — Part-time: Oct 2021 & 2022 — UK

Consultant

- **Imperial College Consultants (ICON)** — Part-time: Mar 2022 – Sep 2022 — UK & USA
  - Research on outlier identification and forecast for UK commercial properties index
- **Ernst & Young (in collaboration with STAI CDT)** — Part-time: Jan 2021 – Apr 2021 — UK
  - Research on back-door poisoning attacks identification for Large Language Models

4most Europe Ltd

- Head of Data Science — Feb 2018 – Sep 2020 — UK & Germany
  - Responsible for the development and delivery of data science projects for UK and German financial institutions
  - Design, prototype and build an interactive ML model monitoring tool (Tech of the Year - National Credit Awards '22)
- Managing Consultant — Jul 2016 – Jan 2018 — UK & Canada
  - Responsible for statistical modelling and analysis of Credit Risk for UK and Canadian financial institutions
  - Responsible for the development and performance of a team of 3 junior consultants
- Credit Risk Consultant — Jul 2015 – Jun 2016 — UK

Credit Risk Analyst — Full-time: May 2014 – Jun 2015

General Electric Capital - UK Home Lending — UK

EDUCATION

PhD in Safe and Trusted Artificial Intelligence — Full-time: Oct 2020 – Feb 2025

Imperial College London — UK

- Thesis: Causal Discovery for Trustworthy Artificial Intelligence
- Funded by UKRI Safe and Trusted Artificial Intelligence Centre for Doctoral Training

MSc in Applied Statistics (Distinction) — Part-time: Oct 2016 – Oct 2018

University of London - Birkbeck College — UK

- Dissertation: Macroeconomic Modelling through VAR and VEC

BSc in Economics of Financial Markets and Institutions (1:1) — Full-time: Oct 2009 – Oct 2012

University of Rome - Tor Vergata — Italy & Spain

- Dissertation: The Tobin Tax between Theoretical Wherewithal and Practical Feasibility

RESEARCH FUNDING

- **Project Lead** | *CausAlly: Collaborative Causal Discovery for Decision Making* | **Imperial Policy Forum** | 2y Fellowship **Awarded**
- **Co-Investigator** | *CArLA: Causal Argumentative Learning Assistant* | **ERC Proof of Concept** | €150,000 **Awarded**

TEACHING & MENTORSHIP

Course Instruction & Guest Lectures

- **Imperial College London** – COMP70076 Ethics, Fairness and Explanation in AI (Guest Lecture, Spring 2026): *Causal Discovery for Trustworthy AI*
- **Luiss Guido Carli, Rome** – BSc Data Science In Action (Guest Lecture, Autumn 2025): *Causal Discovery for Trustworthy AI*
- **Politecnico di Milano & UCL, London** – MSc in Fintech (Guest Lecture, Spring 2022): *From Credit Risk to XAI Research*
- **Imperial College London** – COMP70050 Introduction to Machine Learning (Tutorials and Marking, 2021–2022)

Thesis Supervision

- Sam Waugh (MSc Computing 2026): Causal ABA Learning
- Tiancheng Yang (MSc Computing 2026): Explainable AI for Answer Sets and its application to Causal Discovery
- Dhruv Himatsingka (MEng Computing 2026): Bridging Gradual Semantics and GNNs for Argumentative Causal Discovery
- Zihao Li (MSc Computing 2025): Leveraging Large Language Models for Causal Discovery [9]
- Yeva Hunanyan (MSc Computing 2025): Exploring Semantics for Argumentative Causal Discovery
- Preesha Gehlot (MEng Computing 2025): Graph Neural Networks for Approximate Argumentative Reasoning [8]

PROFESSIONAL SERVICE

Invited Research Talks & Seminars

- *Optimal Correction Sets for Argumentative Causal Discovery*. **AIG Oberseminar, FernUniversität in Hagen**, Germany (2026)
- *Causal AI You Can Challenge: From Data to Decisions in Banking*. **Lloyds Banking Group Summer School**, London UK (2026)
- *CausAlly: Collaborative Causal Discovery for AI-Aided Decision-Making*. **Minds and Machines Workshop, University of Queensland**, Brisbane Australia (2025)
- *Causal Discovery for Trustworthy AI*. **Lloyds Banking Group Summer School**, London UK (2025)
- *Causal Discovery for Contestable and Explainable AI: From Contestable Neural Networks to Transparent Causal Discovery*. **Robert Gordon University**, Aberdeen UK (2024)
- *Credit Risk Modelling: Data and Techniques Used in the UK Banking Industry*. **Danmarks Nationalbank Conference**, Copenhagen Denmark (2019)

Conference & Journal Reviewing

- **Artificial Intelligence Journal**: Reviewer (2026, 2025)
- **KR**: Reviewer (2026)
- **IJCAI-ECAI**: Reviewer (2026, 2023)
- **AAAI**: Reviewer (2027, 2026)
- **NeurIPS**: Reviewer (2025, 2024)
- **ICLR**: Reviewer (2027, 2025)
- **AISTATS**: Reviewer (2024, 2022); Top-Reviewer (2023)
- **ICML**: Reviewer (2022)

Academic Event Organisation

- **XAI Seminar Series @ Imperial**: Co-organiser (2021–present). Grown from 40 to 450+ subscribers.
- **XAI-FIN at ICAIF**: Co-Chair (2025); Assistant Chair (2023, 2024); Workshop co-organiser (2022). 100+ attendees.

PUBLICATIONS

[9] Li Z, **Russo F**. Leveraging Large Language Models for Causal Discovery: a Constraint-based, Argumentation-driven Approach. Proc. of the 42nd Conference on Uncertainty in Artificial Intelligence. pp. 3631–3667. 2026 (CORE: A).

[8] Gehlot P, Rapberger A, **Russo F**, Toni F. Heterogeneous Graph Neural Networks for Assumption-Based Argumentation. Proc. of the 40th AAAI Conference on Artificial Intelligence. pp. 19117–19125. 2026 (CORE: A*).

[7] Rapberger A, **Russo F**, Rago A, Toni F. On Gradual Semantics for Assumption-Based Argumentation. Proc. of the 22nd International Conference on Principles of Knowledge Representation and Reasoning. pp. 512–522. 2025 (CORE: A*).

[6] **Russo F**, Toni F. Shapley-PC: Constraint-based Causal Structure Learning with a Shapley Inspired Framework. Proc. of the 4th Conference on Causal Learning and Reasoning. pp. 292–339. 2025.

[5] **Russo F**, Rapberger A, Toni F. Argumentative Causal Discovery. Proc. of the 21st International Conference on Principles of Knowledge Representation and Reasoning. pp. 938–949. 2024 (CORE: A*).

[4] Leofante F, Ayoobi H, Dejl A, Freedman G, Gorur D, Jiang J, Paulino-Passos G, Rago A, Rapberger A, **Russo F**, Yin X, Zhang D, Toni F. Contestable AI Needs Computational Argumentation. Proc. of the 21st International Conference on Principles of Knowledge Representation and Reasoning. pp. 888–896. 2024 (CORE: A*).

[3] **Russo F**, Toni F. Causal Discovery and Knowledge Injection for Contestable Neural Networks. Proc. of the 26th European Conference on Artificial Intelligence. pp. 2025–2032. 2023 (CORE: A).

[2] **Russo F**. Argumentation for Interactive Causal Discovery. Proc. of the 32nd International Joint Conference on Artificial Intelligence. pp. 7091–7092. 2023 (CORE: A*).

[1] Rago A, **Russo F**, Albini E, Toni F, Baroni P. Explaining Classifiers' Outputs with Causal Models and Argumentation. Journal of Applied Logics – IfCoLog Journal of Logics and their Applications. Vol. 10, No. 3. pp. 421–449. 2023.